

Task: Two Way Conveyor



Project Description

For transporting finished products from the ground level of your factory to the storage that is located on the first floor, you should reprogram an installed conveyor belt. The conveyor belt can transport products from the production (ground level) to the storage (first floor) and down again for shipping (ground level), when needed. Therefore the conveyor belt has got one motor, which can be controlled in up and down direction. A mode selector switch decides whether the system is in manual mode or in automatic mode. In manual mode the conveyor belt will drive packages up, down or stop depending on the position of a three position selector switch.

The conveyor should automatically start transporting a package up, when a photoelectric sensor (NC) recognizes a package on the bottom (ground level). When the capacitive end position sensor on top of the conveyor (first level) recognizes a package, the conveyor stops. The behavior is the same for up and down movement, using the opposing sensors and actuators.

To minimize errors, there is a simple LED system using four LED's, which indicates the current status of the system. Two LED's (one green and one red) are installed on the ground level and two LED's (also green and red) are installed on the first level. If the red LED is active on one level, the system is currently active from the other side and no packages should be placed on the belt. If the green LED is active, packages can be placed. If the green LED flashes with 1 Hz, a package transfer is currently active from your side. The LED stops flashing, when the package has been transferred.

Mode Decision

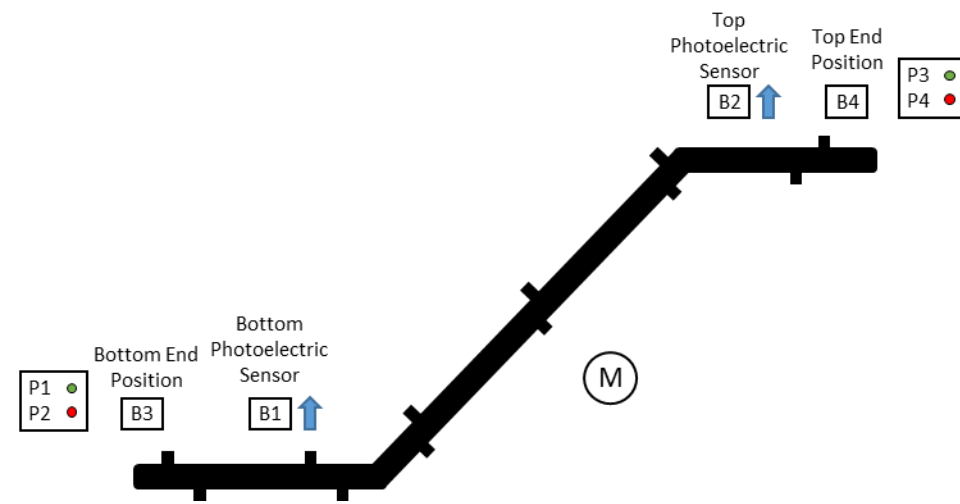
If the two position selector switch S1 on the operator panel is set to “Manual”, the conveyor belt can be controlled by using the three position selector switch S4 to move up, down or stand still.

When S1 is switched to “Automatic”, the system will not longer react to S4 and enter “Pre-Automatic Mode”. In Pre-Automatic Mode, the LED of the start button SP2 flashes with a frequency of 1Hz indicating the system is ready to start. A final acknowledging press of “SP2” will turn on “Automatic Mode” and the LED of SP2 is lit. By pressing the stop button “SP3”, the system will go back to Pre-Automatic Mode.

Automatic Mode Operation

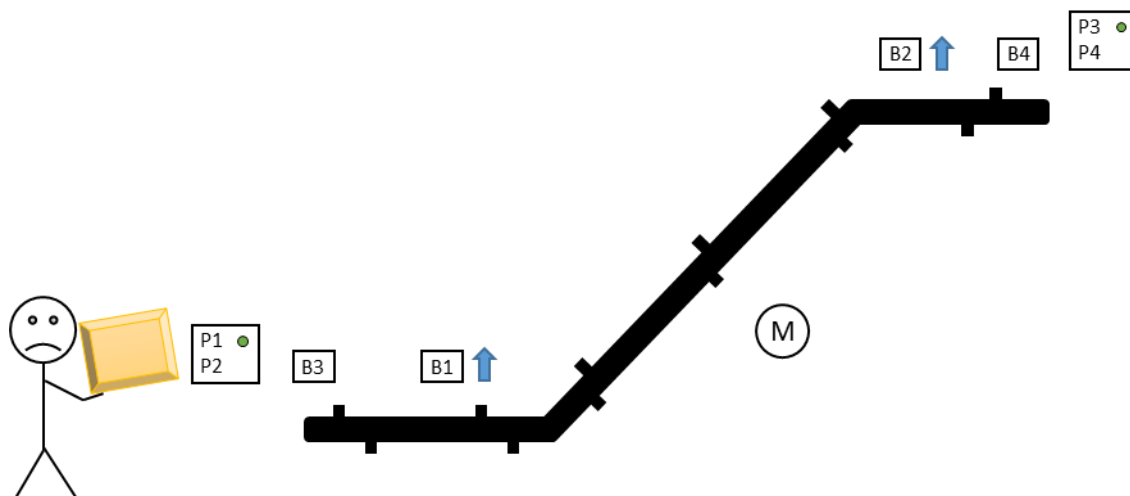
The following illustrations can help you with getting a deeper understanding of the system.

Component Locations:

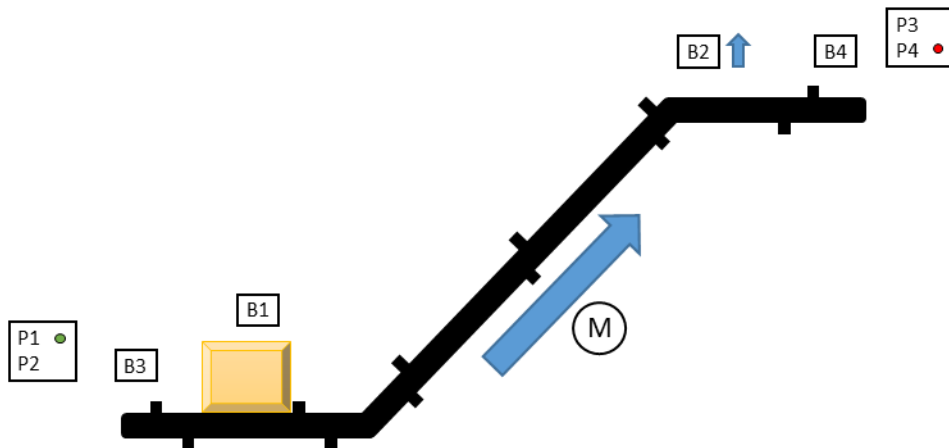


Movement:

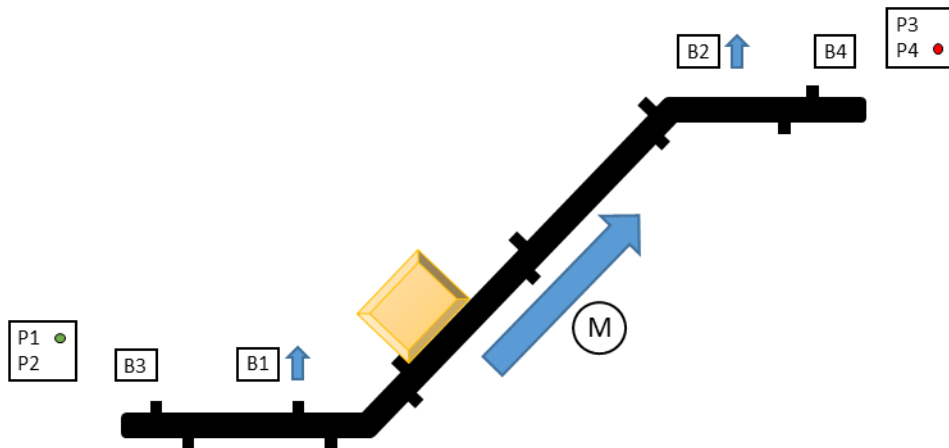
In the beginning, the belt is empty. No sensors are active, the motor and red LED's are turned off. Both green LED's are on, indicating that a package can be places on the top or bottom location.



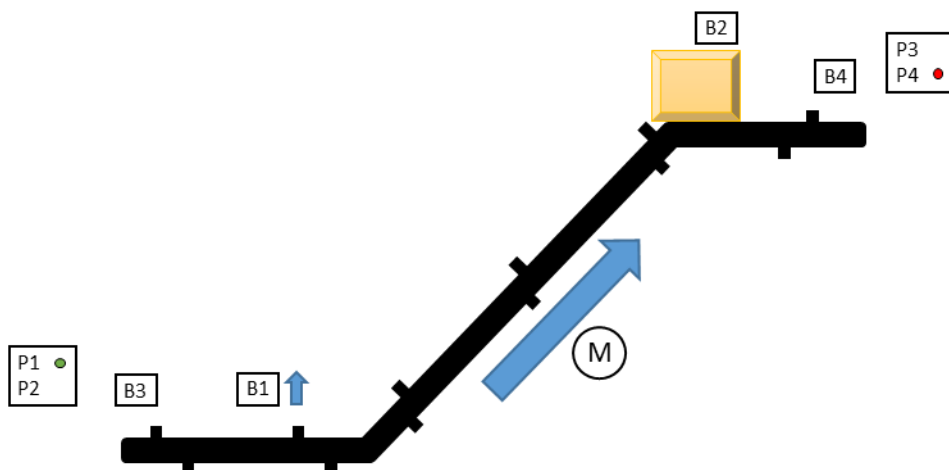
As soon as a package is placed at the location of the normally closed “Photoelectric Sensor Bottom (B1)”, the belt runs in up direction. The red LED on the top turns on and the green LED on top turns off, indicating that no component should be placed from the top. The green LED on the bottom starts flashing with 1 Hz, indicating that a package is on the way up.



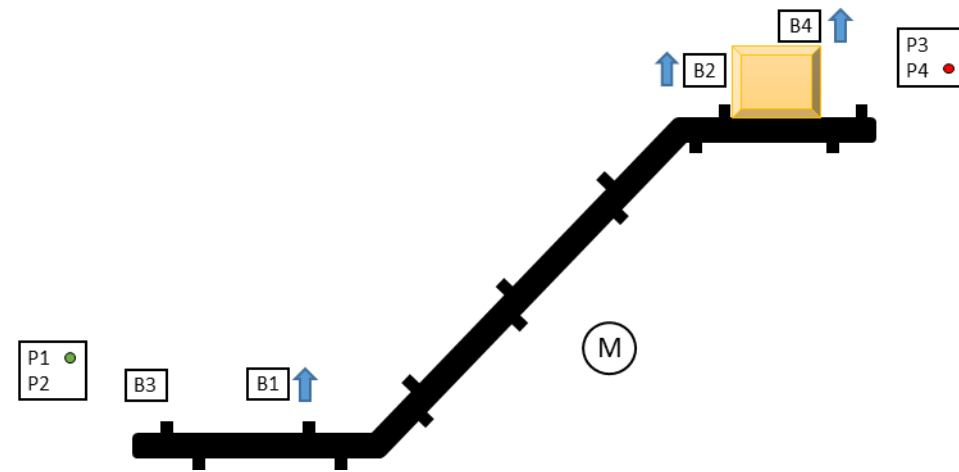
The belt does not stop, when the photoelectric sensor gets deactivated, the motor latches itself.



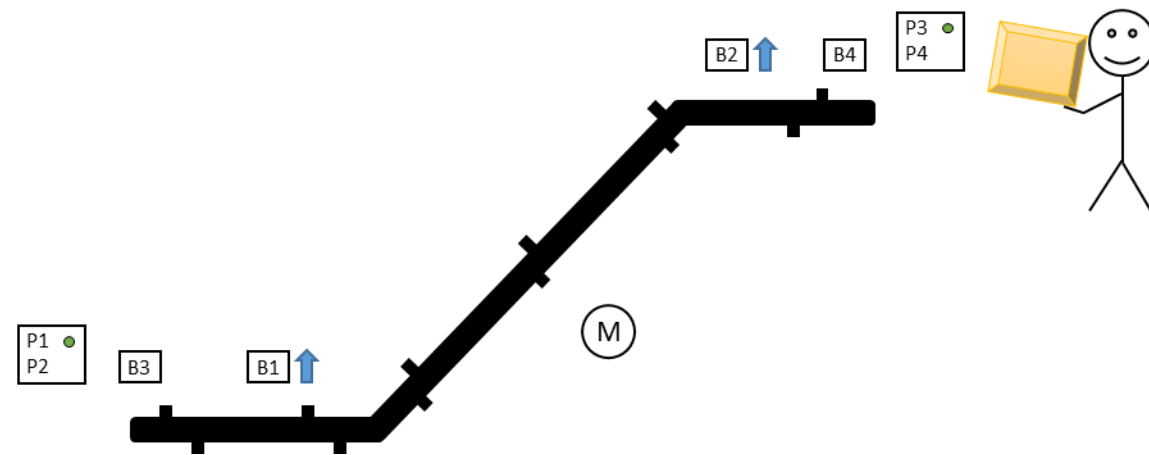
Passing sensor B2 in upwards movement must not influence the motor and LED's.



As soon as the upper end position is reached, indicated by the “End Position Sensor Top (B4)”, the belt stops moving.



The package can now manually be taken off the belt and put in the exact storage location. When the package is taken off, the system is again in its starting state and packages can be placed at both ends.



The system works the same way in both directions, using the opposing sensors and actuators!

Program Organization

No inputs should be read / no outputs should be written in OB1, but only in functions. One function should be named “Motor”, where all outputs to control the up and down movement of the conveyor should be programmed into. Make sure that the conveyor does not move upwards and downwards at the same time. Another function named “LEDs” should be responsible for the four signal LEDs. It should make no difference, in which order those two functions are called in the main. A third function “Modes” should be responsible for the mode decision.

I/O-List

The complete wiring for the system is already done and the signals are wired into the PLC input and output modules as shown below. The information should be added into the PLC tag table.

Name	Address	Comment
Photoelectric Sensor Bottom (NC)	I0.0	-B1:14
End Position Sensor Bottom	I0.1	-B3:14
Photoelectric Sensor Top (NC)	I0.2	-B2:14
End Position Sensor Top	I0.3	-B4:14
Mode Selector Switch	I1.0	-S1:14
Start Button	I1.1	-SP2:14
Stop Button	I1.2	-SP3:14
Manual Up Selector Switch	I1.3	-S4:14
Manual Down Selector Switch	I1.4	-S4:24
Conveyor Up	Q0.0	-K1:A1
Conveyor Down	Q0.1	-K2:A1
LED Start Button	Q0.2	-SP2:X1
LED Stop Button	Q0.3	-SP3:X1
LED Green Bottom	Q0.4	-P1:X1
LED Red Bottom	Q0.5	-P2:X1
LED Green Top	Q0.6	-P3:X1
LED Red Top	Q0.7	-P4:X1

PLC Hardware

The system uses an S7-1500 controller with one digital I/O module installed.

The CPU type is “CPU 1516-3 PN/DP” with the order number “6ES7 516-3AP03-0AB0”, using the newest version.

The I/O module needed is type “DI 16x24VDC / DQ 16x24VDC/0.5A BA” with the order number “6ES7 523-1BL00-0AA0”.

The first I/O module should use input byte 0..1 and output byte 0..1.

Task

Create a program in TIA-Portal for the described system and test your program using PLCSIM

To achieve full points, the program needs to function as described and the program has to be commented and organized in a proper way (i.e. comments, function calls, Tag-Names / Variable-Names,...).